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FOCUS ON TRICUSPID AND MITRAL VALVES

ORIGINAL RESEARCH: STRUCTURAL

Percutaneous Edge-to-Edge Repair for Tricuspid Regurgitation

3-Year Outcomes From the TRILUMINATE Study

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ABSTRACT

BACKGROUND Tricuspid regurgitation (TR) is a common valve disease that has a significant impact on patients' quality of life.

OBJECTIVES This study sought to report the final 3-year outcomes of tricuspid transcatheter edge-to-edge repair (T-TEER) with the TriClip (Abbott) implant from the TRILUMINATE (TRILUMINATE Study With Abbott Transcatheter Clip Repair System in Patients With Moderate or Greater TR) study for the treatment of severe symptomatic TR.

METHODS The TRILUMINATE study (N = 98 subjects) is an international, prospective, single-arm, multicenter study to investigate the safety and performance of T-TEER with the TriClip implant in patients with symptomatic moderate or greater TR. Echocardiographic assessments were performed at a core laboratory.

RESULTS At 3 years, TR was reduced to moderate or less in 79% of subjects, and a reduction of at least 1 grade was achieved in 92% of subjects. TR reduction achieved at 1 year was sustained through 3 years. Subjects also experienced an improvement in heart failure symptoms assessed by NYHA functional class and quality of life assessed by the Kansas City Cardiomyopathy Questionnaire at 3 years compared to baseline. The site-reported heart failure hospitalization rate decreased from 0.56 events/patient-year 1 year before device implantation to 0.14 events/patient-year 3 years after device implantation, representing a reduction of 75% ($P < 0.0001$).

CONCLUSIONS In the longest follow-up to date of any T-TEER therapy, the TRILUMINATE study demonstrated that the TriClip procedure is both safe and effective, with sustained benefits at 3 years in subjects with symptomatic moderate or greater TR. (TRILUMINATE Study With Abbott Transcatheter Clip Repair System in Patients With Moderate or Greater TR [TRILUMINATE]; NCT03227757) (JACC Cardiovasc Interv. 2024;17:2113–2122) © 2024 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

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ABBREVIATIONS AND ACRONYMS

HFH = heart failure hospitalization

KCCQ = Kansas City Cardiomyopathy Questionnaire

TR = tricuspid regurgitation

T-TEER = tricuspid transcatheter edge-to-edge repair

Patients with severe tricuspid regurgitation (TR) are often symptomatic; excessive fatigue and peripheral edema impact their quality of life and oftentimes lead to hospitalizations. Additionally, TR is a common contributor to advanced heart disease and, when left untreated, has been associated with increased morbidity and worse survival.[1] Although medical therapy is available as a treatment for severe TR, it is largely limited to diuretic agents that alone do not correct TR and of which management and compliance can be challenging for patients.[2] Surgical treatment of isolated TR is underused, in part because of the late presentation of the disease and the consequent increased risk of operative mortality, which has been reported to be as high as 8% to 10% in national registries.[3-6] The recognition that TR has a significant impact on patient outcomes and quality of life has bolstered the pursuit of alternative treatment options to surgery or medical therapy alone.

The TriClip tricuspid transcatheter edge-to-edge repair (T-TEER) system (Abbott) is a transvenous-transfemoral percutaneous leaflet repair device for the treatment of TR with a dedicated design for the tricuspid anatomy. The TRILUMINATE (TRILUMINATE Study With Abbott Transcatheter Clip Repair System in Patients With Moderate or Greater TR) study is the first prospective multicenter study conducted in 21 heart valve centers in Europe and the United States to evaluate the safety and performance of the TriClip T-TEER system for the treatment of patients with symptomatic moderate or greater TR, at high risk for tricuspid valve surgery, and with valve anatomies suitable for T-TEER. The primary endpoints of safety (a composite of major adverse events) and performance (TR reduction) of the TRILUMINATE study as well as improvements in clinical status were met at 6 months and at 1 and 2 years, as previously reported.[7-9]

We now report on the final 3-year results of the TRILUMINATE study in which we assess the long-term durability of TR reduction with the TriClip T-TEER system and the resulting impact of TR reduction on right ventricular remodeling, quality of life, and heart failure symptoms.

METHODS

STUDY DESIGN. The TRILUMINATE study is a prospective, single-arm, multicenter study designed to evaluate the safety and performance of the TriClip T-TEER system. A total of 21 sites in Europe and the United States were included. Potential subjects were evaluated for enrollment according to the specified inclusion and exclusion criteria by an eligibility committee, including interventional cardiologists, cardiovascular surgeons, heart failure specialists, and echocardiographers, as described previously.[7] In brief, symptomatic subjects at high surgical risk with moderate or greater TR and no indications for left-sided or pulmonary valve correction were included in this study, whereas subjects with severe pulmonary artery systolic pressure (>60 mm Hg), previous tricuspid valve procedures, or coaptation gaps >10 mm were excluded. The study was approved by local ethics committees and the respective health authorities of the participating countries. All subjects provided written informed consent.

All echocardiograms were analyzed by an independent core laboratory that followed the American Society of Echocardiography standards.[10] TR was assessed using standard 2-dimensional color Doppler methods and was graded using the 5-class grading scheme (mild, moderate, severe, massive, and torrential) to enable a broader and yet differentiated assessment.[11] Among others, parameters used to assess the severity of TR included biplane vena contracta width, effective regurgitant orifice area, and regurgitant volume.

STUDY ENDPOINTS. The primary safety endpoint was a composite of major adverse events at 6 months, including cardiovascular mortality, myocardial infarction, stroke, new-onset renal failure, endocarditis requiring surgery, and nonelective cardiovascular surgery for device-related adverse events postprocedure. The primary efficacy endpoint was a reduction in TR severity by at least 1 grade at 30 days. The performance goal for the primary safety endpoint was 39%, and for the primary efficacy endpoint, it was 35%. Both primary endpoints were met and reported previously.[7] In addition, secondary and descriptive endpoints (post hoc and exploratory) were evaluated

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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to comprehensively assess the performance of the TriClip device. Procedural success was defined as reduction of TR to moderate or less (as assessed by the echocardiography core laboratory) at 30 days. All major adverse events, including cardiovascular mortality, myocardial infarction, stroke, new-onset renal failure, endocarditis requiring surgery, and nonelective cardiovascular surgery for device-related adverse events, were adjudicated by an independent events committee. Additional safety endpoints (eg, major bleeding and new-onset liver failure) and heart failure hospitalizations (HFHs) were assessed at each site according to definitions provided in the clinical investigation plan. Clinical status was assessed using NYHA functional class, the Kansas City Cardiomyopathy Questionnaire score (KCCQ), and the 6-minute walk distance test; all assessments were administered by site investigators.

STATISTICAL ANALYSIS. All analyses were performed on the 98 enrolled subjects who had an attempted procedure upon femoral vein puncture. Outcomes from the first 85 subjects who made up the analysis cohort (ie, subjects were included in the analysis of endpoints) were reported previously.[7-9] Longitudinal data for continuous variables were analyzed using a generalized linear model with autoregressive correlation to account for repeated measures. Data are presented as the least squares mean ± SE for continuous variables. Proportions are presented for categoric variables, and the Bowker or McNemar test was used to compare paired nominal data. Survival analysis was conducted to analyze time-to-event variables. Subjects without events were censored at their last known event-free time point. Survival curves were constructed using Kaplan-Meier estimates. The evaluation of TR reduction to moderate or less at 30 days on the time-to-event variable was analyzed using the Cox proportional hazards model with 1 main effect in the model—TR reduction to moderate or less at 30 days (yes/no). All statistical analyses were performed using SAS version 9.4 (SAS Institute).

RESULTS

BASELINE CHARACTERISTICS. Baseline characteristics are summarized in **Table 1** for all 98 subjects. The mean age was 77.5 ± 8.1 years, and 66% of subjects were female. Subjects had significant comorbidities, including atrial fibrillation (92%), hypertension (86%), renal disease (42%), diabetes (24%), and prior myocardial infarction (18%). Thirty-one percent of subjects had a prior mitral intervention, and 76% were classified as NYHA functional class III/IV. Of the 98 subjects, 25 deaths, 10 withdrawals, and 3 missed visits resulted in 60 subjects with a completed 3-year visit.

TABLE 1 Baseline Characteristics (N = 98)

Age, y	77.5 ± 8.1
Female	66
Atrial fibrillation	92
Arterial hypertension	86
Renal Disease	42
Prior mitral intervention	31
Diabetes	24
Prior myocardial infarction	18
CRT/ICD/PPM	14
NYHA functional class III/IV	76
EuroSCORE II, %	8.3 ± 10.0
Tricuspid regurgitation severity	
Moderate	5.1
Severe	32.7
Massive	28.6
Torrential	33.7
Right ventricular end-diastolic dimension, cm	5.3 ± 0.7
Right ventricular fractional area change, %	36.2 ± 7.3
Tricuspid annular plane systolic excursion, cm	1.5 ± 0.3
Inferior caval vein diameter, cm	2.3 ± 0.6

Values are mean ± SD or %.

CRT = cardiac resynchronization therapy; ICD = implantable cardioverter-defibrillator; PPM, permanent pacemaker.

REDUCTION IN TR. Forty-eight subjects had evaluable TR severity at baseline and the 3-year follow-up. A reduction to moderate or less TR at 3 years was achieved in 79% (38/48) of subjects at 3 years (P = 0.0001, baseline vs 3 years), and, overall, the change in TR grade between 1 year and 3 years (P = 0.912) was not significant (**Central Illustration**, Supplemental Figure 1). At the same time, TR reduction of at least 1 grade was achieved in 92% (44/48) of subjects at 3 years. Significant reductions in vena contracta width (1.7 ± 0.6 to 0.6 ± 0.3 cm; $P < 0.0001$), effective regurgitant orifice area (0.6 ± 0.3 to 0.3 ± 0.1 cm^2; $P < 0.0001$), regurgitant volume (51.3 ± 18.2 to 22.6 ± 10.8 mL/beat; $P < 0.0001$), and proximal isovelocity surface area radius (0.9 ± 0.2 to 0.6 ± 0.2 cm; $P < 0.0001$) were observed between baseline and the 3-year follow-up (**Table 2**).

HEART FAILURE SYMPTOMS. Significant improvements in clinical outcomes were observed for clinical status and quality of life assessment. The proportion of subjects classified as NYHA functional class III/IV decreased from 76% of subjects at baseline to 19% at

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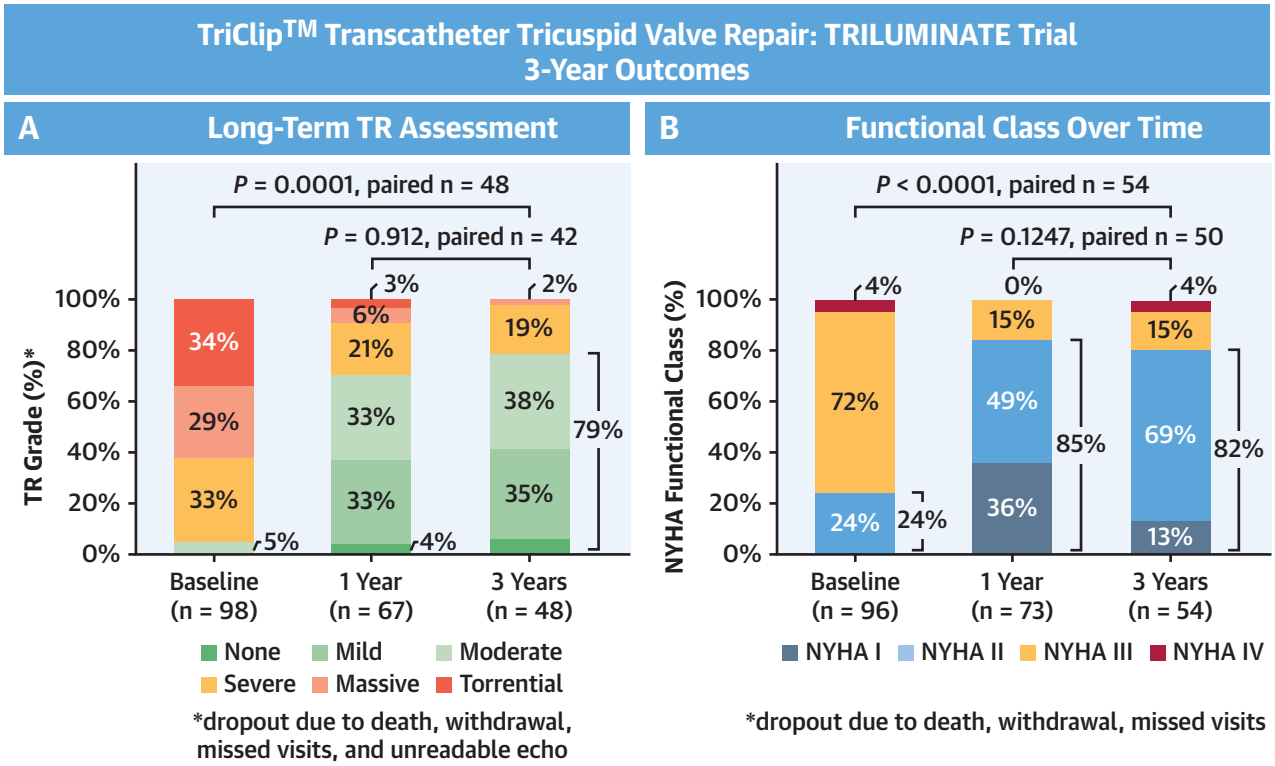
CENTRAL ILLUSTRATION **Tricuspid Regurgitation Repair Through 3 Years**



- At 3-year follow-up, there was a substantial and durable reduction in TR following TEER
- NYHA functional class improved following TEER, and this benefit remained over time

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TEER = transcatheter edge-to-edge repair; TR = tricuspid regurgitation.

3 years ($P < 0.0001$, **Central Illustration**). No significant change in NYHA functional class occurred between the 1- and 3-year follow-ups. Self-assessed heart failure symptoms measured by the KCCQ for subjects with KCCQ scores at baseline and 1 and 3 years is shown in **Figure 1B**. The KCCQ score decreased after 3 years from the improvement noted at 1 year (−12 ± 21 points; $P = 0.0001$); however, a mean improvement of 10 ± 3 points from baseline to the 3-year follow-up remained ($P = 0.006$).

STRUCTURAL AND HEMODYNAMIC CHANGES. Sustained remodeling of the right heart over 3 years was evident through the decrease in the right ventricular end-diastolic dimension (5.3 ± 0.1 cm at baseline to 4.9 ± 0.1 cm at the 3-year follow-up; $P < 0.0001$) and no significant change from the 1-year follow-up (**Figure 2**). The right atrial volume was not significantly different from baseline at the 3-year follow-up (127.5 ± 5.4 mL to 118.5 ± 7.9 mL; $P = 0.195$). The tricuspid annular diameter also remained significantly decreased from baseline to 3 years (4.3 ± 0.6 cm to 4.1 ± 0.6 cm; $P = 0.0145$), and the reduction at 1 year was sustained at 3 years (4.0 ± 0.6 cm to 4.1 ± 0.6 cm; $P = 0.0872$). Fractional area change was mildly reduced from 36.9% ± 7.3% at baseline to 34.6% ± 6.6% at the 3-year follow-up ($P = 0.038$) (**Table 2**). The improvement in tricuspid annular plane systolic excursion at the 1-year follow-up did not remain at 3 years.[7,9] There was also a significant decrease in left ventricular function at 3 years (59.7 ± 7.6 to 54.9 ± 7.0; $P < 0.0001$), with a nonsignificant reduction in forward stroke volume (60.7 ± 14.3 to 57.7 ± 15.8; $P = 0.30$).

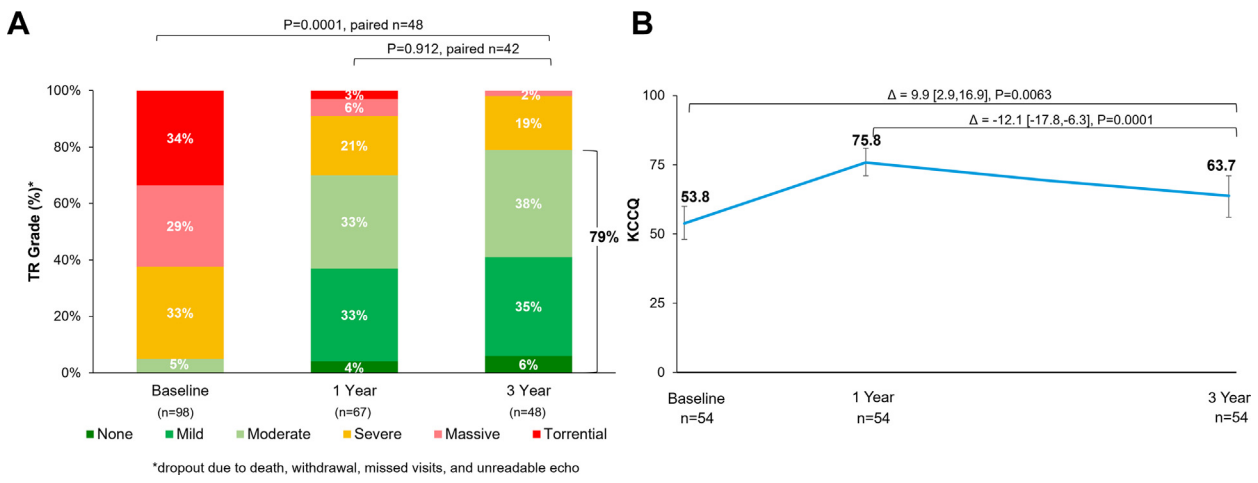
TABLE 2 Echocardiographic Endpoints

	Median (Q1-Q3)			*P* Value Baseline vs 3 Years[a]
	Baseline	1 Year	3 Years	
Tricuspid regurgitation				
Effective regurgitant orifice area, cm^2	0.61 (0.46-0.75) (76)	0.27 (0.16-0.44) (39)	0.23 (0.19-0.37)	<0.0001
Regurgitant volume, mL/beat	52.90 (38.86-62.50) (76)	26.00 (12.40-40.10) (39)	21.85 (13.90-31.00) (30)	<0.0001
Regurgitant jet area, cm^2	13.45 (10.30-17.10) (85)	5.65 (3.66-9.66) (60)	5.50 (3.90-8.33) (32)	<0.0001
Vena contracta width, cm	1.59 (1.20-2.07) (97)	0.68 (0.42-1.00) (67)	0.50 (0.37-0.74) (47)	<0.0001
PISA radius, cm	0.90 (0.80-1.00) (76)	0.60 (0.45-0.80) (39)	0.60 (0.40-0.70) (31)	<0.0001
IVC diameter, cm	2.30 (2.00-2.70) (76)	2.10 (1.74-2.43) (58)	1.95 (1.67-2.30) (42)	0.0526
Forward stroke volume (left ventricle), mL	56.90 (51.00-73.00) (74)	65.40 (53.00-74.00) (57)	54.00 (44.80-63.60) (39)	0.3322
Left ventricular ejection fraction, %	61.10 (57.00-63.40) (86)	60.00 (54.00-64.60) (65)	56.00 (49.90-60.00) (47)	<0.0001
RV end-diastolic dimension (base), cm	5.20 (4.70-5.70) (95)	4.70 (4.25-5.30) (64)	4.65 (4.30-5.00) (50)	<0.0001
Tricuspid annular diameter, cm	4.23 (3.96-4.70) (97)	4.10 (3.60-4.50) (65)	3.90 (3.70-4.60) (49)	0.0145
Right atrial volume, mL	112.80 (92.80-154.20) (90)	107.00 (85.40-143.40) (65)	105.30 (66.60-145.10) (49)	0.0442
RV fractional area change, %	37.30 (31.30-40.20) (80)	39.00 (36.40-41.80) (58)	34.10 (30.30-40.00) (46)	0.0381
RV systolic pressure, mm Hg	40.69 (35.40-50.05) (71)	38.43 (33.15-46.81) (43)	40.75 (33.54-46.72) (36)	0.2635
TAPSE, cm	1.50 (1.24-1.70) (92)	1.60 (1.40-1.80) (67)	1.40 (1.10-1.60) (50)	0.0968
RV global longitudinal strain, %	−14.80 (−17.00 to −11.50) (57)	−15.47 (−17.70 to −12.60) (17)	−15.32 (−18.95 to −13.30) (36)	0.0742

Values are median (Q1-Q3) (n). [a]*P* value for subjects with both baseline and 3-year data.
IVC = inferior vena cava; PISA = proximal isovelocity surface area; RV = right ventricular; TAPSE = tricuspid annular plane systolic excursion.

SAFETY OUTCOMES AND HEART FAILURE IMPROVEMENT. Major adverse events occurred in 25% (24/96) of subjects, of which only 1 event (new-onset renal failure) occurred within 30 days. Major adverse events included cardiovascular mortality (n = 18), myocardial infarction (n = 1), stroke (n = 4), and new onset of renal failure (n = 8) (Table 3). There were no cases of endocarditis requiring surgery or nonelective cardiovascular surgery for a device-related adverse event. All-cause mortality was 27%, with 25 deaths through 3 years. Other clinical safety endpoints through the 3-year follow-up include major bleeding (10%, 10/96), stroke (4%, 4/96), new-onset liver failure of (1%, 1/96), and new pacemaker

FIGURE 1 TR Reduction and KCCQ Improvement



(A) Tricuspid regurgitation (TR) reduction sustained from 1 year to 3 years. (B) The decrease in the Kansas City Cardiomyopathy Questionnaire (KCCQ) score observed from 1 year to 3 years, although a significant improvement from baseline to 3 years remains.

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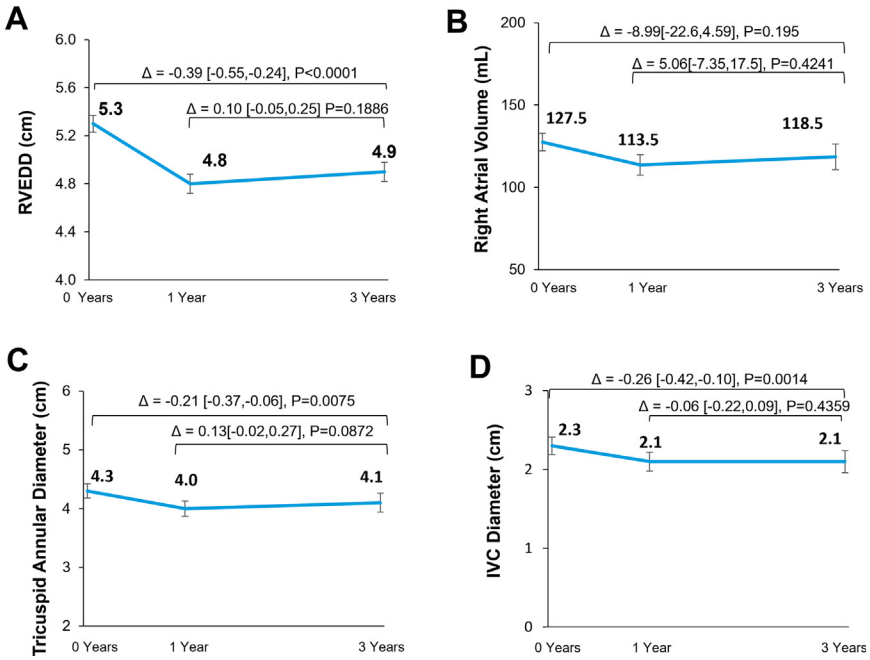
FIGURE 2 Right Heart Remodeling



(A) The right ventricular end-diastolic dimension (RVEDD) decreased significantly at 3 years compared with baseline. (B) The right atrial volume fluctuated throughout the 3-year follow-up. (C) The tricuspid annulus diameter decreased significantly at 3 years compared with baseline, and (D) the inferior vena cava (IVC) diameter decreased significantly at 3 years compared with baseline.

implantation (2%, 2/96). There were no instances of pulmonary thromboembolism, evidence of thrombosis, or embolization.

The site-reported HFH rate decreased from 0.56 events/patient-year 1 year before device implantation to 0.14 events/patient-year 3 years after the TriClip procedure, thus representing a reduction of 75% (Figure 3). This reduction was not attributed to changes in patient management because diuretic usage over the duration of follow-up remained the same in 87.3% (55/63), decreased in 7.9% (5/63), and increased in 4.8% (3/63) of subjects. Additionally, the all-cause hospitalization rate decreased from 1.34 events/patient-year 1 year before device implantation (site-reported data) to 0.63 events/patient-year 3 years after the procedure.

Subjects achieving a TR reduction to moderate or less at 30 days were less likely to experience HFH or death at 3 years compared with subjects who remained with severe or greater TR at 30 days (30.7% vs 50.7%; HR: 0.48; 95% CI: 0.25-0.93; $P = 0.026$) (Figure 4).

TABLE 3 Safety Events

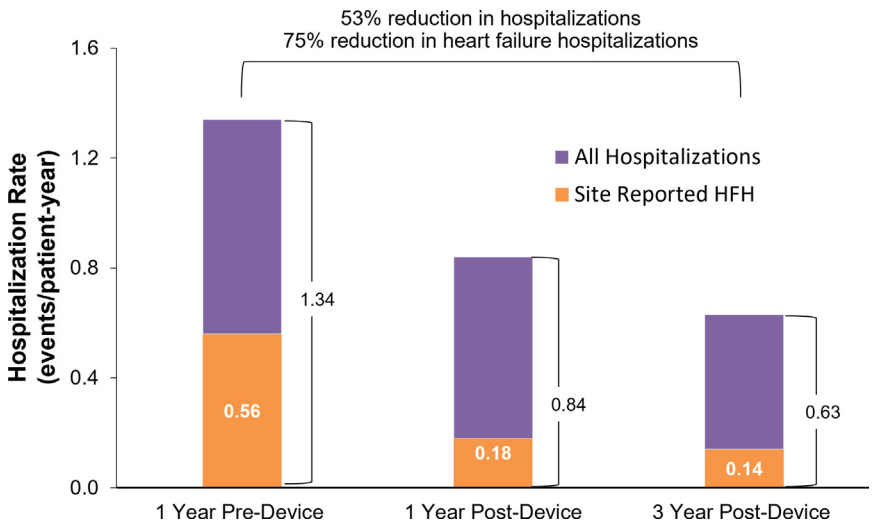
	30 Days	3 Years
Major adverse events	1.0 (1)	25.0 (24)
Cardiovascular mortality	0.0 (0)	18.8 (18)
Myocardial infarction	0.0 (0)	1.0 (1)
Stroke	0.0 (0)	4.2 (4)
New-onset renal failure	1.0 (1)	8.3 (8)
Endocarditis requiring surgery	0.0 (0)	0.0 (0)
Nonelective cardiovascular surgery for device-related adverse event	0.0 (0)	0.0 (0)

Values are % (n).

DISCUSSION

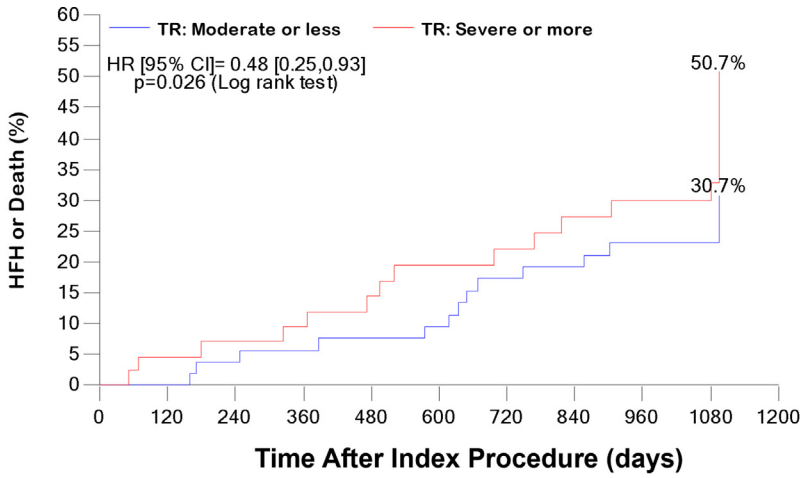
The final 3-year results of the TRILUMINATE study demonstrate that the TriClip device reliably reduced TR to moderate or less and sustained this reduction in the majority of subjects over 3 years. Subjects also had durable favorable remodeling of the right heart and, as a possible consequence, experienced improvement in heart failure symptoms assessed by NYHA functional class and quality of life assessed by the KCCQ at 3 years compared with baseline. As the longest follow-up to

FIGURE 3 Hospitalizations per Patient-Year



A 53% reduction in all-cause hospitalizations compared with site-reported hospitalizations 1 year before device implantation was observed with a 75% reduction in site-reported heart failure hospitalization (HFH).

FIGURE 4 HFH or Death After 3 Years by TR Reduction Observed at 30 Days



	0	30	180	365	730	1095
TR: Moderate or less						
# At Risk	54	54	51	48	43	24
% Event Rate	0.0%	0.0%	3.8%	7.6%	17.2%	30.7%
TR: Severe or more						
# At Risk	44	43	34	32	26	15
% Event Rate	0.0%	2.3%	18.6%	23.4%	37.7%	50.7%

When subjects were stratified by 30-day TR of moderate or less or severe or more, a significant decrease in HFH or death was observed in subjects with 30-day moderate or less TR ($P = 0.026$, log-rank test). Abbreviations as in Figures 1 and 4.

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date of any T-TEER therapy, the following results are crucial for the understanding of T-TEER therapy with the TriClip device as a treatment for severe TR:

1. First, safety of the procedure in multimorbid and often frail patients is encouraging.[7,8] There were minimal major adverse events periprocedurally, after 30 days, and after 3 years.
2. Second, in most patients, TR was meaningfully reduced by T-TEER, and this effect was sustained over 3 years.
3. Third, after an initial improvement in the KCCQ score through 1 year, there was a decline at 3 years; however, more than a 10-point improvement remained over baseline.
4. Fourth, the amount of TR reduction may be related to outcomes. Although a control group was not included in this study design and the number of subjects remaining in the study at 3 years was small, patients who had TR reduced to moderate or less had fewer events compared with patients with less relevant TR reduction.

This latter finding has been shown in the Transcatheter Tricuspid Valve Therapies (TriValve) registry, and the results of the TRILUMINATE Pivotal randomized controlled trial (RCT) have also shown that the amount of TR reduction is associated with greater improvements in the KCCQ score.[2] At 3 years, it is unsurprising that the improvement in the KCCQ score was decreased in comparison to the improvement realized at the 1-year follow-up. This decline in the KCCQ score could possibly be explained by seemingly progressive cardiac dysfunction given the presence of a decrease in both forward stroke volume and left ventricular ejection fraction in this elderly, comorbid population. Importantly, TR reduction was sustained through 3 years, which suggests that the reduction in the KCCQ is not associated with an increase in TR grade; however, this association and KCCQ decline is better investigated in studies with a control arm, such as TRILUMINATE Pivotal, because historic data on KCCQ decline over time in subjects with severe or greater TR are not readily available. Nevertheless, the 10-point improvement in the KCCQ score at 3 years is still reassuring.

The data presented herein are even more encouraging because only the small TriClip NT (Abbott) device was used and not the currently available TriClip G4 System (Abbott), with both longer and wider TriClip XT and XTW clips, as well as independent leaflet grasping functionality to optimize clip placement. The limited clip availability in TRILUMINATE may be 1 explanation as to why the TR reduction in the recently published TRILUMINATE Pivotal RCT and bRIGHT (An Observational Real-World Study Evaluating Severe Tricuspid Regurgitation Patients Treated With the Abbott TriClip Device) was even more efficacious because the NT device is rarely used in contemporary practice.[2,12] However, procedural differences such as increased implanter experience and advances in imaging techniques could have also contributed to the difference in efficacy of TR reduction between trials. The application of these novel iterations of T-TEER devices, in addition to advancements in implanter experience and imaging capabilities, may allow for the treatment of patients with more complex diseases in the future. Additionally, the treatment of more complex patients may be also relevant for the impact of transcatheter edge-to-edge repair on patient outcomes.

Currently, it remains unclear if the TR reduction observed here directly causes favorable outcomes; the presented data are derived from an uncontrolled study, and the 1-year results of the TRILUMINATE Pivotal RCT were neutral with respect to the effect of TriClip treatment on HFH and mortality.[2] In the TRILUMINATE Pivotal RCT, it is tempting to speculate that the follow-up in the TRILUMINATE Pivotal RCT was too short or that the patients were just too healthy to benefit from T-TEER during the time of the reported follow-up given the significant reduction in HFH compared to site-reported events before treatment. For comparison, a smaller proportion of patients in the TRILUMINATE Pivotal trial had kidney and liver disease, fewer were NYHA functional class III/IV, and fewer experienced HFH in the year before TriClip implantation (Supplemental Table 1). Furthermore, the HFH event rate was unexpectedly low in the control arm of the TRILUMINATE Pivotal RCT, making a distinct significant improvement in the device group challenging. Additional analyses are needed to further understand the progression of tricuspid disease and the role that concomitant disease processes play in long-term patient outcomes. However, our encouraging 3-years results justify and warrant these further controlled studies on T-TEER in patients with severe, symptomatic TR, particularly in light of the favorable safety profile and the lack of other treatment options.

STUDY LIMITATIONS. The TRILUMINATE study was not a randomized controlled study, and, therefore, we cannot establish a direct relationship between the reduction of TR and clinical endpoints. Additionally, the heart valve centers that enrolled subjects in the study may reflect a higher experience level, although experience before this study has been limited to off-label MitraClip (Abbott) device use. The study was

also conducted with early device iterations, and learning curves could have an effect in some centers. Although the TriClip device is currently available in multiple sizes, only the smaller NT device was used in this study. The 4 different sizes of clip in the latest device generation, the TriClip G4 System, in addition to the improved maneuverability and ability to grasp leaflets independently, could potentially serve a wider range of patients compared with the results reported in this analysis. Finally, assessments of the KCCQ, NYHA functional class, and the 6-minute walk test were unblinded and site or investigator assessed and may be affected by awareness bias.

CONCLUSIONS

The final report of the outcomes from the prospective single-arm international TRILUMINATE study demonstrates that the TriClip repair system is safe, and TR reduction is sustained through 3 years. TR reduction to moderate or less at 3 years was not significantly different from the reduction at 1 year ($P = 0.912$). Subjects also experienced fewer heart failure symptoms as demonstrated by the 75% reduction in HFH at 3-years postprocedure compared to 1 year prior and a sustained improvement in self-assessed heart failure symptoms through the KCCQ.

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PERSPECTIVES

WHAT IS KNOWN? The primary endpoints of safety (a composite of major adverse events) and performance (TR reduction) of the TRILUMINATE study as well as improvements in clinical status were met at 6 months and at 1 and 2 years, as previously reported.

WHAT IS NEW? The 3-year outcomes demonstrate that subjects with symptomatic moderate or greater TR treated with the TriClip show sustained benefits, including TR reduction and quality of life improvements at 3 years.

WHAT IS NEXT? In the longest follow-up to date of any T-TEER therapy, the TRILUMINATE study demonstrated that the TriClip procedure is both safe and effective, with sustained benefits at 3 years in subjects with symptomatic moderate or greater TR.

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KEY WORDS leaflet repair, TEER, TriClip, tricuspid regurgitation, tricuspid repair

APPENDIX For a supplemental table and figure, please see the online version of this paper.